

Algebra & vector

Objective:-

(i) The value of $\begin{vmatrix} 3 & 4 \\ -2 & 8 \end{vmatrix}$ is -

Ans:- $\Delta = 24 - (-8) = 32.$

(ii) The value of x in $\begin{vmatrix} x & 1 \\ 2 & 4 \end{vmatrix} = 0$ is -

Ans: $4x - 2 = 0$
 $x = \frac{2}{4} = \frac{1}{2}.$

(iii) The value of x if $\begin{vmatrix} x & 3 \\ 2 & 1 \end{vmatrix} = \begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix}$ is -

Ans: $x - 6 = 8 - 3$
 $\Rightarrow x - 6 = 5$
 $\therefore x = 11.$

(iv) The value of $\begin{vmatrix} 1 & -1 \\ y & x \end{vmatrix}$ is -

Ans: $x - (-y) = x + y.$

(v) If a matrix has 12 elements, how many such order are formed?

Ans: $12 = 1 \times 12, 12 \times 1, 2 \times 6, 6 \times 2, 3 \times 4, 4 \times 3$ Total = 6.

(vi) If A is an orthogonal then the value of AA' is -

Ans: $I.$

(vii) The value of $(A')'$ is -

Ans: $A.$

(viii) The value of $|Adj(A)|$ is -

Ans: $|A|^{n-1}$

(ix) The value of $(AB)^T$ is

Ans: $B^T A^T.$

(x) If A & B are two symmetric matrices then $(AB - BA)$ is -

Ans: skew symmetric matrix.

(xi) If A is a square matrix and I is also a matrix of A then

AI is -

Ans: A

(xii) If I is a unit matrix then

I^n is -

Ans: I

(xiii) If $\vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}$ then

$|\vec{a}|$ is -

Ans: $|\vec{a}| = \sqrt{2^2 + 2^2 + 1^2} = 3.$

(xiv) If $\vec{a}$ be any vector and $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ then unit vector $\hat{a}$ is

Ans: $\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{\sqrt{3}} (\hat{i} + \hat{j} + \hat{k}).$

(xv) The value of $\hat{i} \cdot \hat{i} + \hat{j} \cdot \hat{j} + \hat{k} \cdot \hat{k}$ is -

Ans: 3

(xvi) The value of $\hat{i} \cdot (\hat{j} \times \hat{k}) + \hat{j} \cdot (\hat{k} \times \hat{i}) + \hat{k} \cdot (\hat{i} \times \hat{j})$ is -

Ans: $\hat{i} \cdot \hat{i} + \hat{j} \cdot \hat{j} + \hat{k} \cdot \hat{k} = 3.$

(xvii) The value of $(\hat{i} \times \hat{i}) + (\hat{j} \times \hat{j}) + (\hat{k} \times \hat{k})$ is -

Ans: $0 + 0 + 0 = 0.$

(xviii) If $\vec{a} \perp \vec{b}$ then $\vec{a} \cdot \vec{b}$ is -

Ans: 0

(xix) If $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ & $\vec{b} = \lambda\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{a} \perp \vec{b}$ then λ is -

Ans: $\because \vec{a} \perp \vec{b} \therefore \vec{a} \cdot \vec{b} = 0$
 $1 + 4 + 1 = 0$
 $1 = -5.$

SUBJECTIVE

1. Prove that $\begin{vmatrix} 1+a^2+b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3$

2. Prove that $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$

3. Prove that $\begin{vmatrix} x+4 & 2x & 2x \\ 2x & x+4 & 2x \\ 2x & 2x & x+4 \end{vmatrix} = (5x+4)(x-4)^2$

4. Prove that $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$

5. Prove that $\begin{vmatrix} a-b-c & 2a & 2b \\ 2b & b-c-a & 2c \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$

6. Using Cramer's rule solve the following system equations

(i) $x+2y+z=1$, $x-y+2z=-1$, $2x+y-z=2$

(ii) $x-y=2$, $2x+y-z=3$, $y-2z=1$

7. Construct a matrix of order 3×4 such that $a_{ij} = 2i+j$

8. Find the matrix 'X' & 'Y' such that

$X+Y = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$ & $2X-Y = \begin{bmatrix} 1 & 6 \\ -1 & 2 \end{bmatrix}$

9. If $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$ & $B = \begin{bmatrix} 4 & -1 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ find AB.

Is 'BA' possible also find BA.

10. If $A = \begin{bmatrix} 2 & 1 & -3 \\ 1 & -1 & 4 \\ 2 & 1 & 5 \end{bmatrix}$ find A^T .

11. If $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \\ 2 & -1 \end{bmatrix}$ & $B = \begin{bmatrix} 3 & 4 \\ 1 & 7 \\ 2 & 1 \end{bmatrix}$ prove that

$(A+B)' = A' + B'$

12. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ prove that

$A^2 - 4A + 5I = O$, Also find A^{-1} .

13. Using matrix method solve the following system of equations

$x+2y+z=2$

$2x-y+3z=1$

$x-y+z=-1$

14. Using matrix method solve the following system of equations

$x-2y=1$

$y+3z=-1$

$x+y+z=2$

15. If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$

prove that A is an orthogonal matrix.

16. If $A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 4 \\ 5 & -1 & 3 \end{bmatrix}$

express it in the sum of symmetric & skew symmetric matrices.

Subjective VECTOR

1. Prove that the vectors $3\hat{i} + 5\hat{j} + 2\hat{k}$, $2\hat{i} - 3\hat{j} - 5\hat{k}$ & $5\hat{i} + 2\hat{j} - 3\hat{k}$ form the side of an equilateral triangle.
2. Prove that three points $\hat{i} - 2\hat{j} - 8\hat{k}$, $5\hat{i} - 2\hat{k}$ & $11\hat{i} + 3\hat{j} + 7\hat{k}$ are collinear.
3. Prove that the vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ & $3\hat{i} - 4\hat{j} - 4\hat{k}$ form a right angled triangle.
4. Find the direction cosines of a vector $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$
5. Prove that the vector $\vec{r} = \hat{i} + \hat{j} + \hat{k}$ is equally inclined to the co-ordinates axes.
6. Find a unit vector perpendicular to each of the vectors $\vec{a}$ & $\vec{b}$ where $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ & $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$
7. Show that the vectors $\hat{i} - 3\hat{j} + 2\hat{k}$, $2\hat{i} - 4\hat{j} - \hat{k}$ & $3\hat{i} + 2\hat{j} - \hat{k}$ are linearly independent.
8. If $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$ & $\vec{b} = \hat{i} + 3\hat{j} - 2\hat{k}$ find $|\vec{a} \times \vec{b}|$.
9. If $\vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}$ & $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$ find the angle between $\vec{a}$ & $\vec{b}$.
10. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$ find the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$.
11. If $\vec{a} + \vec{b} + \vec{c} = 0$ prove that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} \neq \vec{c} \times \vec{a}$.
12. Prove that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) + \vec{c} \times (\vec{a} + \vec{b}) = 0$.
13. If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$ & $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$ find the projection of (i) $\vec{a}$ on $\vec{b}$ (ii) $\vec{b}$ on $\vec{a}$.
14. Find the area of a triangle if its two sides are represented by vectors $2\hat{i} + \hat{j} + \hat{k}$ & $\hat{i} + 2\hat{j} - \hat{k}$.
15. Find the area of a parallelogram if its two adjacent sides are represented by vectors $2\hat{i} + 2\hat{j} + \hat{k}$ & $\hat{i} - 2\hat{j} + 3\hat{k}$.

Objective

UNIT-2 Differential Calculus

* Function:-

(i) If $f(x) = \log \frac{1+x}{1-x}$ then $f\left(\frac{2x}{1+x^2}\right)$ is equal to -

Ans: $2f(x)$.

(ii) If $f(x) = x + \frac{1}{x}$, then the value of $f(x) - f\left(\frac{1}{x}\right)$ is -

Ans: 0

(iii) If $f(x) = \log x$ the $f(x) + f\left(\frac{1}{x}\right)$ is equal to -

Ans: 0

(iv) The domain of function $y = \frac{x}{1+x^2}$ is -

Ans: $\mathbb{R}$ or $]-\infty, +\infty[$ or $(-\infty, \infty)$

(v) The domain of function $f(x) = y = \frac{1}{\sqrt{x+2}}$ is -

Ans: $]-2, \infty[$ or $(-2, \infty)$

(vi) The domain of the function $f(x) = \sqrt{(x-2)(x-3)}$ is -

Ans: $]-\infty, 2] \cup [3, \infty[$ or $]-\infty, 2] \cup [3, \infty)$.

(vii) Find the domain of $f(x) = \frac{x}{x^2 - 3x + 2}$

Ans: $\mathbb{R} - \{1, 2\}$

(viii) The range of the function $y = \frac{x}{1+x^2}$ is equal to -

Ans: $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(ix) The range of $f(x) = \sin x$ is -

Ans: $[-1, 1]$

(x) The range of $f(x) = \frac{1}{2 - \sin 3x}$ is -

(xi) The range of $f(x) = \sqrt{9-x^2}$ is $[0, 3]$

Ans: $\left[\frac{1}{3}, 1\right]$.

* Limit:-

(i) The value of $\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta}$ is -

Ans: 1

(ii) The value of $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$ is -

Ans: 3

(iii) The value of $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ is -

Ans: $\frac{1}{180}$.

(iv) The value of $\lim_{x \rightarrow 2} \frac{x^5 - 32}{x - 2}$ is -

Ans: 80.

(v) The value of $\lim_{x \rightarrow 0} \frac{\sin 5x}{\tan 2x}$ is -

Ans: $\frac{5}{2}$.

(vi) The value of $\lim_{x \rightarrow 1} \frac{x^n - 1}{x - 1}$ is -

Ans: n.

(vii) The value of $\lim_{x \rightarrow 2} \frac{x^5 - 32}{x^2 - 4}$ is -

Ans: 20.

(viii) The value of $\lim_{x \rightarrow 1} \frac{x-1}{x^{\frac{1}{4}} - 1}$ is -

Ans: 4.

(ix) The value of $\lim_{x \rightarrow 9} \frac{x^n - 9^n}{x - 9}$ is -

Ans: $n 9^{n-1}$.

(x) $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$ is -

Ans: 1

(xi) $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x}$ is -

Ans: 1

(xii) $\lim_{x \rightarrow 0} \frac{\sin^2 x}{x}$ is -

Ans: 0

Differentiation

(i) Differential coefficient of $\sec x$ with respect to x is -

Ans: $\frac{1}{\sqrt{x^2 - 1}}$

(ii) Differential coefficient of $\cos 2x$ is -

Ans: $-2 \sin 2x$

(iii) Differential coefficient of e^{3x} is -

Ans: $3e^{3x}$.

(iv) Differential coefficient of $\sin x$ w.r.t x is - (xvii) If $y = 4x^3 - 2x^2 - 2x + 7$ then value of $\frac{d^2y}{dx^2}$ at $x = 2$ is -
Ans: $\sec^3 x$.

(v) Differential coefficient of a^x w.r.t x is -

Ans: $a^x \log a$

(vi) Differential coefficient of 2^x w.r.t x is -

Ans: $2^x \log 2$

(vii) If $y = x^x$ then $\frac{dy}{dx}$ is equal to

Ans: $x^x (1 + \log x)$.

(viii) Differential coefficient of e^x w.r.t x is -

Ans: $a e^{ax}$.

(ix) Differential coefficient of $\tan^{-1} x$ w.r.t x is -

Ans: $\frac{1}{1+x^2}$.

(x) Differential coefficient of $\sqrt{\cos x}$ w.r.t x is -

Ans: $\frac{-\sin x}{2\sqrt{\cos x}}$.

(xi) Differential coefficient of $\log(\log x)$ w.r.t x is -

Ans: $\frac{1}{x \log x}$.

(xii) Differential coefficient of $\sec x$ w.r.t x is -

Ans: $\sin x$.

(xiii) If $x = ae^{2t}$, $y = be^{3t}$ then $\frac{dy}{dx}$ is -

Ans: $-\frac{a}{b} e^{2t}$.

(xiv) Differential coefficient of $\sqrt{\tan x}$ w.r.t x is -

Ans: $\frac{\sec^2 x}{2\sqrt{\tan x}}$.

(xv) If $x = a \cos \theta$, $y = b \sin \theta$ then value of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{4}$ is -

Ans: -1 .

(xvi) If $y = x^x$ then $\frac{dy}{dx}$ is equal to -

Ans: $x^x (1 + \log x)$.

(xvii) If $y = 4x^3 - 2x^2 - 2x + 7$ then value of $\frac{d^2y}{dx^2}$ at $x = 2$ is -

Ans: 38.

(xviii) The function $f(x) =$

$x^3 - 2x^2 + x + 6$ will be Max or Min when

Ans: $3x^2 - 4x + 1 = 0$

(xix) The equation of tangent at $(1, 6)$ to the curve

$y = 2x^2 + 3x + 1$

Ans: $7x - y - 1 = 0$.

(xx) If $y = x^3 + 4x^2 + 5$ then value of $\frac{d^2y}{dx^2}$ at $x = 0$ is -

Ans: 8

(xxi) If $y = x^2 + x + 1$ then value of $\frac{dy}{dx}$ at $x = 1$ is

Ans: $\frac{dy}{dx} = 2x + 1 = 3$

(xxii) If $y = x^3 + x + 1$ find the slope of

tangent at $x = -1$

Ans: $\frac{dy}{dx} = 3x^2 + 1 = 4$

(xxiii) If $y = x^2 + x + 1$ find the slope of

normal at $x = 1$

Ans: $-\frac{1}{3}$

(xxiv) For Max value of $y = f(x)$ the

value of $\frac{d^2y}{dx^2}$ is

Ans: negative

(xxv) For Min value of $y = f(x)$ the value of

$\frac{d^2y}{dx^2}$ is

Ans: positive.

LIMIT

UNIT-2

SUBJECTIVE

$$1. \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+2x} - \sqrt{1-3x}}$$

$$2. \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x} - \sqrt{1-x}}$$

$$3. \lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \sqrt{1-x^2}}{x^2}$$

$$4. \lim_{x \rightarrow a} \frac{x^{p/2} - a^{p/2}}{x^{3/2} - a^{3/2}}$$

$$5. \lim_{x \rightarrow 0} x(3 \operatorname{cosec} 2x - 4 \cot 3x)$$

$$6. \lim_{x \rightarrow y} \frac{\sin x - \sin y}{x - y}$$

$$7. \lim_{x \rightarrow y} \frac{\tan x - \tan y}{x - y}$$

$$8. \lim_{x \rightarrow \alpha} \frac{x \sin \alpha - \alpha \sin x}{x - \alpha}$$

$$9. \lim_{n \rightarrow \infty} \frac{1^2+1}{n^3} + \frac{2^2+2}{n^3} + \frac{3^2+3}{n^3} + \dots + \frac{n^2+n}{n^3}$$

$$10. \lim_{x \rightarrow -3} \frac{x^5 + 243}{x^3 + 27}$$

$$11. \lim_{x \rightarrow \infty} \frac{ax^p + bx^{p-3} + c}{dx^2 + ex^{q-2} + k} \text{ where } p, q > 0$$

and a, b, c, d, e, k are constants.

$$5. \text{ Test the continuity of a function } f(x) \text{ at } x=0 \text{ where } f(x) = \begin{cases} \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$6. \text{ If the function } f(x) \text{ is continuous at } x=1 \text{ find the value of 'a' \& 'b' where } f(x) = \begin{cases} 3x+b & x > 1 \\ 5ax-2b & x < 1 \\ 11 & x = 1 \end{cases}$$

$$7. \text{ Test the continuity of a function } f(x) \text{ at } x=0 \text{ where } f(x) = \begin{cases} x^2 \sin \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$8. \text{ Find the value of a and b for the function } f(x) \text{ which is continuous at } x=0 \text{ where}$$

$$f(x) = \begin{cases} \frac{a \sin 3x}{x} & x > 0 \\ b \left(\frac{\sqrt{1+x}-1}{x} \right) & x < 0 \\ \frac{3}{2} & x = 0 \end{cases}$$

$$9. \text{ Test the continuity of a function } f(x) \text{ at } x=1 \text{ where } f(x) = |x-1|$$

CONTINUITY

$$1. \text{ Test the continuity of a function } f(x) \text{ at } x=0 \text{ where } f(x) = |x|$$

$$2. \text{ Test the continuity of a function } f(x) \text{ at } x=a \text{ where } f(x) = \begin{cases} \frac{x^2-a^2}{x-a} & x \neq a \\ 2a & x = a \end{cases}$$

$$3. \text{ Test the continuity of a function } f(x) \text{ at } x=0 \text{ where}$$

$$f(x) = \begin{cases} \frac{\sin x}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}$$

$$4. \text{ Test the continuity of a function } f(x) \text{ at } x=0 \text{ where } f(x) = \begin{cases} 2x-1 & x < 0 \\ 2x+1 & x \geq 0 \end{cases}$$

$$10. \text{ Test the continuity of a function } f(x) \text{ at } x=1$$

$$\text{where } f(x) = \begin{cases} \frac{x^2-3x+2}{x-1} & x > 1 \\ 1-2x & x < 1 \\ -1 & x = 1 \end{cases}$$

DIFFERENTIATION

APPLICATION OF DERIVATIVE

UNIT-3

Q1. Find d.c of the following *
from the 1st principle

- (i) x^n (ii) $\tan x$ (iii) $\sqrt{\tan x}$
(iv) a^x (v) e^{ax} (vi) $\sqrt{\sin x}$
(vii) $\log x$

$\frac{dy}{dx}$ as rate measurer:-

1. If the radius of a circle increases with a rate of 6 cm/sec, find the rate of change in its area if the radius is 50 cm.

Q2. Find $\frac{dy}{dx}$ of the following:

- (i) $x^m y^n = (x-y)^{m+n}$
(ii) $x^4 y^3 = (x-y)^7$
(iii) $y = \sin\{\cos(\tan\sqrt{x})\}$
(iv) $y = \log(x + \sqrt{x^2 - a^2})$
(v) $y = \sin\{\log(\tan x)\}$
(vi) $y = \log(x^2 \sqrt{x^2 + 1})$
(vii) $x = at^2, y = 2at$
(viii) $x = a(t + \sin t), y = a(1 - \cos t)$

Find $\frac{dy}{dx}$ at $t = \frac{\pi}{2}$

(ix) If $x = a(\theta - \sin \theta), y = a(1 - \cos \theta)$

Find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{2}$

(x) $y = x^x + (\sin x)^x$

(xi) $y = (\sin x)^{\cos x} + (\cos x)^{\sin x}$

Q3. (i) If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

Find $\frac{dy}{dx}$

(ii) If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$ prove

that $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{1-x^2}$

(iii) If $x\sqrt{1+y} + y\sqrt{1+x} = 0$ prove that

$\frac{dy}{dx} = \frac{-1}{(1+x)^2}$

(iv) If $y = x^{x^{x^{x^{\dots}}}}$, find $\frac{dy}{dx}$

(v) If $x^y + y^x = 2$, find $\frac{dy}{dx}$

(vi) If $y = (\tan^{-1} x)^2$ prove that $(1+x^2)^2 y_2 + 2x(1+x^2) y_1 = 2$

2. The radius of a sphere increases with a rate of 7 cm/sec find the rate of its surface area if the radius is 35 cm.

3. If the radius of a circle increases 3 cm to 3.01 cm. Find the increase in area.

4. The sides of an equilateral triangle increases with a rate of $3\sqrt{3}$ cm/sec find the rate of change in its area if the side is $12\sqrt{3}$ cm.

5. A man 6 ft tall walks away along a st. line from the foot of a light post 24 ft high at the rate of 3 miles/hour. How fast does the shadow move.

6. If the area of a circle increases with a constant rate prove that its surface circumference increases inversely to its radius.

7. If the radius of a sphere increases with a rate of 3 cm/sec find the rate of change in its volume if its radius is 30 cm.

$$(1+x^2)^2 y_2 + 2x(1+x^2) y_1 = 2$$

TANGENT & NORMAL

Unit-3

1. Find the equation of tangent and normal to the curve $x^3 = y^2$ at a point $(1, 1)$
2. Find the equation of tangent and normal to the curve $y = 5x^2 - 2x + 3$ at a point $(1, 6)$
3. Find the equation of tangent and normal to the curve $y = x^2 + 4x + 1$ at the point where $x = 3$
4. Find the equation of tangent and normal to the curve $y'' = x^2 - 2x + 1$ if the tangent is parallel to the line $4x - 2y + 1 = 0$
5. Find the equation of tangent and normal to the curve $y^2 = 49x$ at a point $(at^2, 2at)$.

* ROLLE'S THEOREM & MEAN VALUE THEOREM

1. verify Rolle's Theorem for the function $f(x) = \sin x$ in $[0, \pi]$
2. verify Rolle's Theorem for the function $f(x) = x^2$ in $[-1, 1]$
3. verify Rolle's Theorem for the function $f(x) = x^2 - 3x + 2$ in $[1, 2]$
4. verify Rolle's Theorem for the function $f(x) = \sin x + \cos x$ in $[0, \frac{\pi}{2}]$
5. verify Lagrange mean value theorem for $f(x) = x^2 + x - 1$ in $[0, 4]$
6. verify Lagrange mean value theorem for $f(x) = x^3 + x^2 - 6x$ in $[-1, 4]$
7. verify Lagrange mean value theorem for $f(x) = \log x$ in $[1, e]$

* MAXIMUM & MINIMUM VALUE OF A FUNCTION

1. Find the maximum and minimum value of a function $f(x) = 2x^3 - 15x^2 + 36x + 11$.
2. Find the minimum value of $y = x^4$
3. Find the maximum & minimum value of $f(x) = x^3 - 6x^2 + 9x + 15$
4. Find the maximum & minimum value of $f(x) = 2x^3 - \frac{25}{2}x^2 - 9x + 1$
5. prove that the maximum value of $(\frac{1}{x})^x$ is $e^{1/e}$
6. prove that the maximum & min value of $f(x) = x^3 - 6x^2 + 9x + 15$ are 19 & 15. respectively.
7. Find the absolute maximum & min value of a function given by $f(x) = 2x^3 - 15x^2 + 36x + 1$ on the interval $[1, 5]$

RADIUS OF CURVATURE

- Q1. Find the radius of curvature at $(1, -1)$ on the curve $y = x^2 - 3x + 1$
- Q2. Find the radius of curvature at the point (x, y) on the curve $y^2 = 4ax$
- Q3. Find the radius of curvature at $(\frac{3a}{2}, \frac{3a}{2})$ on the curve $x^2 + y^2 = 3axy$
- Q4. Find the radius of curvature at any point on the curve
 $x = a(1 + \sin \theta), y = a(1 + \cos \theta)$
- Q5. Find the radius of curvature at any point of the curve $r = ae^{m\theta}$
- Q6. Find the radius of curvature at any point of the curve $r = a(1 + \cos \theta)$
- Q7. Find the radius of curvature of the curve (i) $r^3 = 2ap^2$ (ii) $p^2 = ar$

NOTE (1) Radius of curvature for Cartesian eqⁿ of curve

$$\rho = \frac{\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{3/2}}{\frac{d^2y}{dx^2}}, \quad \frac{d^2y}{dx^2} \neq 0 \quad \text{or} \quad \frac{(1 + y'^2)^{3/2}}{y''}$$

(2) Radius of curvature for parametric eqⁿ of curve

$$\rho = \frac{(x'^2 + y'^2)^{3/2}}{x'y'' - y'x''} \quad \text{where} \quad \begin{aligned} x' &= \frac{dx}{dt}, & x'' &= \frac{d^2x}{dt^2} \\ y' &= \frac{dy}{dt}, & y'' &= \frac{d^2y}{dt^2} \end{aligned}$$

(3) Radius of curvature for pedal eqⁿ of curve

$$\rho = r \frac{dr}{dp}$$

(4) Radius of curvature for polar eqⁿ of curve

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2}$$

Objective types questions

(i) Distance from the origin to a point $(3, 4)$ is -

Ans: 5.

(ii) Distance from the origin to a point $(a \cos \theta, a \sin \theta)$ is -

Ans: $|a|$, or a .

(iii) Distance from the x-axis to a point $(3, 4)$ is -

Ans: 4

(iv) Distance from the y-axis to a point $(2, 3)$ is -

Ans: 2.

(v) In what ratio does x-axis divide a line-segment joining two points $(3, 4)$ & $(2, -3)$ is -

Ans: 4:3

(vi) In what ratio does y-axis divide a line-segment joining two points $(-1, 2)$ & $(3, 4)$ is -

Ans: 1:3.

(vii) The slope and angle of the line passing through two points $(2, 3)$ & $(3, 4)$ are

Ans: 1, 45°

(viii) Find the slope of a line if it passes through two points $(0, 0)$ & $(3, 4)$ is -

Ans: $m = \frac{4-0}{3-0} = \frac{4}{3}$.

(ix) The slope of a line $2x - 3y + 4 = 0$

Ans: $m = \frac{2}{3}$

(x) If two lines are parallel to each other then

Ans: $m_1 = m_2$

(xi) If two lines are $\perp$ to each other then

Ans: $m_1 m_2 = -1$

(xii) If two lines $x + 2y + 3 = 0$ & $4x + y + 7 = 0$ are parallel to each other then

Ans: 1 = 8.

(xiii) If two lines $kx + 4y + 11 = 0$ & $2x + 3y + 7 = 0$ are $\perp$ to each other then k is -

Ans: $k = -6$.

(xiv) Perpendicular distance from the origin to the line $3x + 4y + 5 = 0$ is

Ans: 1

(xv) Perpendicular distance from a point $(1, 1)$ to the line $5x + 12y + 26 = 0$ is

Ans: $d = \frac{43}{13}$.

(xvi) The intercept between the axes by the line $2x + 3y - 6 = 0$ is

Ans: $\sqrt{13}$

(xvii) The centre of a circle $x^2 + y^2 + 2x + 2y - 1 = 0$ is

Ans: $(-1, -1)$.

(xviii) The radius of a circle $x^2 + y^2 + 6x + 8y - 24 = 0$ is

Ans: 7

(xi) The focus of a parabola $y^2 = 8x$ is -

Ans: (2, 0)

(xii) The major axis of an ellipse

$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \quad \text{is -}$$

Ans: 6

(xiii) The minor axis of an ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 \quad \text{is -}$$

Ans: 6

(xiv) Latus rectum of an ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{is -}$$

Ans: $\frac{2b^2}{a}$

(xv) The eccentricity of an

ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is -

Ans: $\frac{\sqrt{5}}{3}$

(xvi) The transverse axis & conjugate axis of the hyperbola

$$x^2 - 2y^2 = 16 \quad \text{are}$$

Ans: Transverse axis = 8
conjugate axis = $4\sqrt{2}$.

SUBJECTIVE

1. Find the equation of a line which passes through a point (1, 2) and it makes an angle of 45° with the line $x - 2y + 7 = 0$.

2. Find the equation of a line which passes through two points (1, 2) & (5, 6).

3. Find the equation of a line which subtends an angle of 60° with x-axis and it cuts an intercept of 4 units along positive side of x-axis.

4. Find the equation of a line which passes through a point (1, 1) and it cuts equal intercept along the co-ordinate axes.

5. Find the equation of a line which bisects the portion intercepted by a line at (2, 3).

6. Find the angles between the lines $x - 2y + 7 = 0$ & $x - 3y + 11 = 0$.

7. Find the equation of legs of an isosceles right angle if its right angle is at (1, 2) and the equation of the hypotenuse is $x + 3y + 1 = 0$.

8. Find the equation of a line which passes through a point (2, 3) and it is parallel to a line $4x + y + 7 = 0$.

9. Find the equation of a line which passes through a point (-1, 2) and it is perpendicular to the line $3x + 5y + 6 = 0$.

10. Find the equation of a perpendicular bisector of a line-segment joining two points (1, 3) & (7, -5).

11. Find the equation of a line which passes through the point of intersection of two lines $x + 2y + 3 = 0$ & $4x + y + 7 = 0$ and a point (3, 4).

OBJECTIVE

12. Find the equation of a line which passes through the point of intersection of two lines $x+2y+3=0$ and $3x+2y+5=0$ and it is parallel to a line $7x+2y+1=0$.
13. Find the equation of a line which passes through the point of intersection of two lines $2x+y-1=0$ & $x-y-2=0$ and it is perpendicular to the line $3x+5y+2=0$.
14. Find the perpendicular distance from a point which is the intersection of two lines $x-y+2=0$ & $2x+y+1=0$ to the line $3x+4y+20=0$.
15. Find the distance between two parallel lines $3x+4y+20=0$ and $3x+4y+35=0$.
16. Find the equation of a circle whose radius is 5 and the equation of two diameters are $2x+y+7=0$ & $x-y+2=0$.
17. Find the centre and radius of a circle $x^2+y^2+6x+8y-24=0$.
18. Find the centre and radius of a circle $2x^2+2y^2+2x+2y+1=0$.
19. Find the equation of a circle which passes through three points $(0,0)$, $(1,0)$ & $(0,1)$.
20. Find the equation of a circle which passes through three points $(1,2)$, $(2,3)$ & $(-1,4)$.
21. Find the equation of a circle which passes through two points $(1,2)$ & $(-1,1)$ and its centre lies on a line $3x+2y+4=0$.
22. Find the equation of a circle which passes through the origin and it cuts ~~intercepts~~ intercepts of 3 & 4 units along x -axis & y -axis respectively.
23. Find the equation of a parabola whose focus is $(2,0)$ and its vertex at the origin.
24. Find the equation of a parabola whose focus is $(1,2)$ and direction is $2x+3y+1=0$.

25. Find the focus, directrix, Latus rectum & vertex of the parabola
 $y^2 + 2x = 4x + 3$

26. Find the equation of the parabola whose focus is $(1, -1)$ and vertex is $(2, 1)$

27. Find the equation of the parabola whose vertex is $(6, -3)$ and directrix is $3x - 5y + 1 = 0$.

28. Find the equation of the parabola whose focus is $(1, 1)$ and the tangent at the vertex is $x + y = 1$.

29. Find the equation of the ellipse, whose focus, directrix and eccentricity are respectively $(-1, 1)$, $x - y + 3 = 0$ and $\frac{1}{2}$.

30. Find the centre, major axis, minor axis, eccentricity, directrices and Latus rectum of the ellipse $4x^2 + y^2 - 8x + 2y + 1 = 0$.

31. Find the equation of the ellipse whose foci are $(2, 3)$ & $(-2, 3)$ and whose semi-minor axis is $\sqrt{5}$.

32. Find the equation of the ellipse whose axes are along the co-ordinate axes, foci at $(0, \pm 4)$ and eccentricity $\frac{4}{5}$.

33. Find the eqⁿ of the ellipse if the distance between the foci is 8 and eccentricity is $\frac{2}{3}$.

34. Find the equation of ellipse if it passes through $(2, 1)$ and its eccentricity is $\frac{1}{2}$.

35. Find the equation of the ellipse whose latus rectum is 4 and its major axis is 8.

36. Find the centre, foci, directrices, Latus rectum, eccentricity, transverse axis & conjugate axis of the hyperbola

(i) $\frac{x^2}{9} - \frac{y^2}{4} = 1$ (ii) $2x^2 - 4x - y^2 + 6y - 17 = 0$

37. Find the equation of the hyperbola for which eccentricity is 2, one focus is $(2, 2)$ and the corresponding directrix is $x + y = 9$.

38. Find the equation of the hyperbola having foci $(\pm 4, 0)$ and latus rectum is 12.

39. Find the equation of the hyperbola having foci $(0, \pm 4)$ & transverse axis is '6'.

OBJECTIVE!.

UNIT-5

STATISTICS & PROBABILITY

Statistics

1. A student obtains marks in 6 subjects as 18, 25, 41, 18, 31, 35, what is his average marks

Ans: - 28.

2. what is the median of the data

3, 9, 2, 8, 7, 1, 1, 4, 5, 6, 6, 2, 9, 7, 8, 4, 6

Ans: 6.

3. what is the mode of the numbers

2, 5, 7, 3, 3, 3, 6

Ans: 3.

4. the relation between, Mean, Median & Mode is

Ans: $\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}$

5. the probability of the occurrence of 6 in throwing a die is -

Ans: $\frac{1}{6}$

6. what is the probability of throwing more than 4 in a single throw from an ordinary die?

Ans: $\frac{1}{3}$.

7. A bag contains 3 white and 2 black balls. the probability of drawing a white ball at random is

Ans: $\frac{3}{5}$.

8. A card is drawn from a pack of 52 playing cards find the probability that the card is a Jack

Ans: $\frac{1}{13}$.

9. on a leap year the probability of having 53 sundays is

Ans: $\frac{2}{7}$

10. on a non leap year the probability of having 53 sundays is

Ans: $\frac{1}{7}$.

11. if $P(A) = \frac{3}{8}$, $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{4}$. then $P(A|B)$ is -

Ans: $\frac{5}{8}$

12. if $P(A \cap B) = \frac{7}{10}$ and $P(B) = \frac{17}{20}$ then $P(A|B)$ is

Ans: $\frac{14}{17}$.

13. if $P(A) = \frac{4}{5}$ and $P(A \cap B) = \frac{7}{10}$ then $P(B|A)$ is

Ans: $P(B|A) = \frac{\frac{7}{10}}{\frac{4}{5}} = \frac{7}{10} \times \frac{5}{4} = \frac{7}{8}$.

14. the mean of 5 observation $x, x+2, x+4, x+6$ and $x+8$ is 11, then the value of x is -

Ans: 7

15. the mean deviation of the data 3, 10, 10, 4, 7, 10, 5 from the mean is

Ans: 2.57.

16. the range of a distribution 2, 5, 9, 1, 7, 11, 15 is -

Ans: $\text{Range} = \text{upper value of the data} - \text{Lower value of the data}$
 $= 15 - 1$
 $= 14$

17. the class mark or class mean of 10-20 is -

Ans: class mark or class mean = 15

18. the frequency of 3 in the data 1, 1, 2, 3, 3, 3, 4, 5, 6, 3 is -

Ans: 4.

SUBJECTIVE

1. Find the mean, median and mode of the following data:-

(i)

Class	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	2	4	5	3	9	2

(ii)

Class	11-15	16-20	21-25	26-30	31-35	36-40	41-45
Frequency	7	10	13	26	55	22	11

2. Find the standard deviation of the following data:-

Class	4-6	6-8	8-10	10-12	12-14	14-16
Frequency	13	10	9	5	8	5

3. Find the variance and S.D of the following data:-

C.I	0-10	10-20	20-30	30-40	40-50
f	2	5	1	3	4

4. Two dice are thrown at random. Find the probability of getting doublets.

5. If two cards are drawn from a pack of 52 playing cards what is the chance or probability that both cards are king.

6. 4 cards are drawn at random find the probability that two cards are king.

7. In a class there are 10 boys and 5 girls. 3 students are to be selected from the class, what is the probability that 2 boys and 1 girl is selected.

8. A bag contains 4 white, 5 black & 3 red balls. A ball is drawn at random, find the probability that the ball is red.

9. Three coins are tossed together, find the probability of getting exactly two heads.

10. 4 cards are drawn from a pack of 52 cards, find the probability that all cards are of different suits

11. The probability of an event is $\frac{3}{7}$. Find odds against the event

12. If $P(A) = \frac{3}{8}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$ then find (i) $P(A \cup B)$ (ii) $P(\bar{A} \cap \bar{B})$

12. The probability of a horse A winning a race is $\frac{1}{5}$ and the probability of a horse B winning the same race is $\frac{1}{4}$. What is the probability that neither of them will win (ii) none of them will win.

14. From a set of 17 cards are numbered 1 to 17, one card is drawn at random, find the probability of getting numbers divisible by 3 or 7.

15. The probability of solving a problem by three students are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ find the probability that the problem will be solved.

16. A man speaks 80% truth and another 90% truth. While stating the same fact what is the probability that they are (i) contradict (ii) agree

17. If $P(A) = \frac{1}{5}$, $P(B) = \frac{3}{5}$, $P(A \cup B) = \frac{3}{4}$ then find (i) $P(A \cap B)$ (ii) $P(\bar{A} \cap \bar{B})$.

17. A die is thrown. Find the probability of the occurrence of a number less than 5 if it is known that only odd nos occur.